

Module 13

Standard Costing

Learning Objectives

By the end of this module you should understand:

1. The relationship between budgeting and standard costing
2. How standard costs are set
3. The motivational aspects of standards
4. The role of flexible budgets
5. How to calculate the principal variances surrounding material, labor and overhead costs
6. The possible reasons for variances and when to investigate them
7. How to calculate sales variances and the managerial significance of the numbers produced

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13.1 Introduction

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13.3 A Word about Motivation

13.4 Flexible Budgets

13.5 The Anatomy of Variances: Materials and Labor

13.6 Responsibility for Variances

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Case Studies

As we've covered in module 12, budgeting is concerned with planning the activities of an organization for a specific time period –usually 12 months

Budgeting

- A plan of action, usually expressed in numbers and amounts, which sets targets for individuals and for the business over a defined future timeframe
- Management must give thought to the future in order to plan to meet its threats and opportunities

Module 12

Sets out the procedures for constructing the budget

Module 13

Considers the mechanism available to management to monitor the actual performance against the plan

We can see from the Go- Straight Trolley Co. case that the budget was ultimately expressed in total numbers, quarter by quarter

Master Budget provides:

**A picture of
Cash Flows**

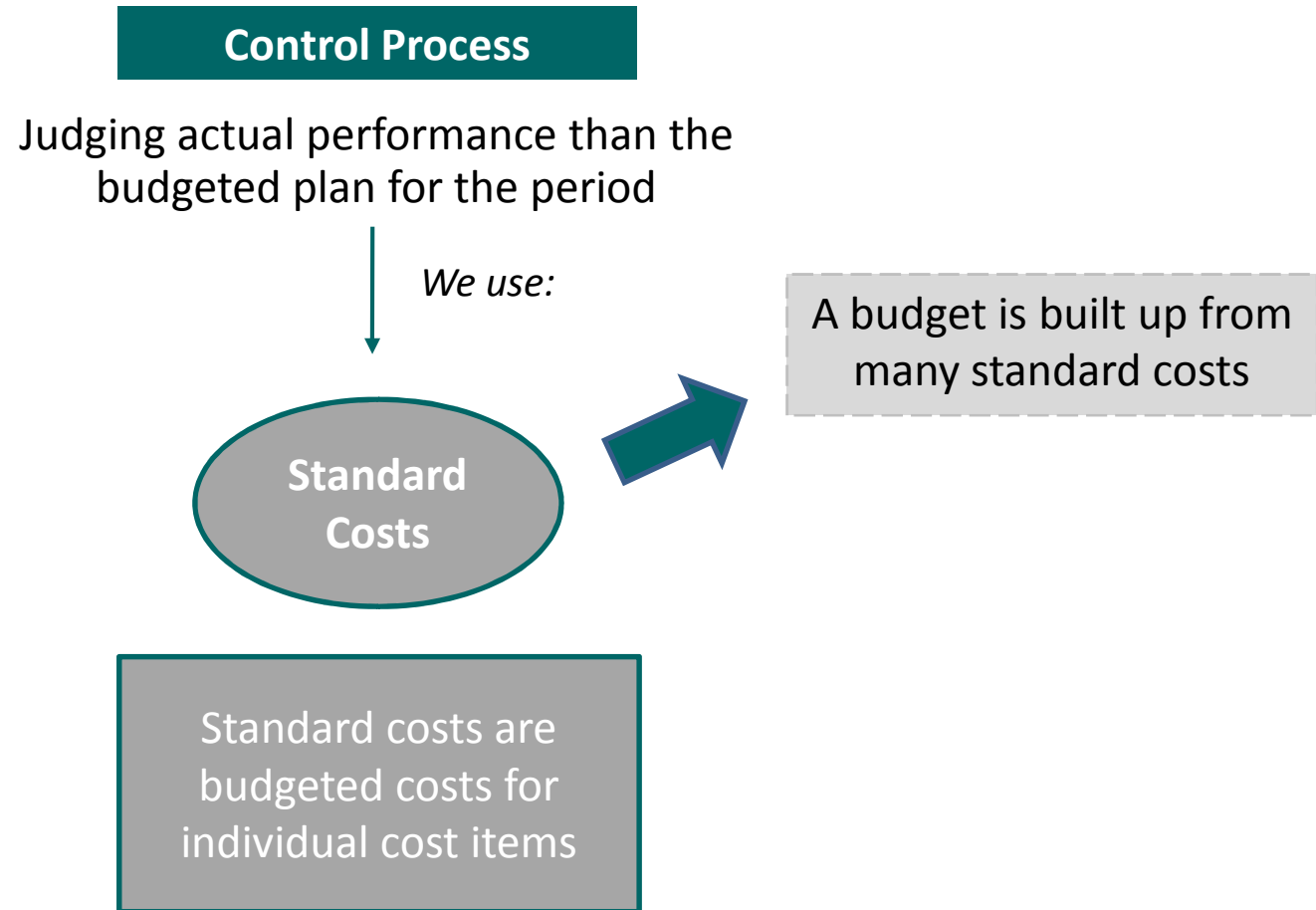
Allows management to gain a forward picture of cash flow; and consequently allows them to negotiate with their bankers in a planned manner

**Answers to
Investment
Analysts**

Equip management to deal with professional investment analysts – who advise potential investors on the company's prospects

Once constructed the budget should be used to control the actual operations of the business for the accounting period under review

There can be no better benchmark to judge actual performance than the budgeted plan for the period – provided that plan was carefully worked out



Standard costs are compared with actual costs & explanations are sought for any differences between the two; where if performance is not up to the desired level remedial managerial action

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Remember the cost profiles of the timber-shelved DVD cabinet and the supermarket trolley

From Module 10		From Module 12	
Standard full cost of DVD cabinet		Standard variable cost of trolley	
	€		€
Direct material		Direct material	
Chipboard	10.00	10 metres of metal	
		@ €1 per metre	10
Veneer	3.00	Direct labour	
		1.5 hours	
Metal trim	2.00	@ €20 per hour	30
	<u>15.00</u>		
		Variable overhead	
		1.5 hours @ €10 per hour	15
Direct labour		Standard variable cost	<u>55</u>
1.5 hours @ €5 per hour	<u>7.50</u>		
Prime cost	<u>22.50</u>		
Manufacturing overhead	<u>2.50</u>		
Manufacturing cost	<u>25.00</u>		
Share of non-manufacturing overhead	<u>5.00</u>		
Standard full cost	<u><u>30.00</u></u>		

Each of these products was costed in some detail with the DVD cabinet being allocated a share of the manufacturing and non-manufacturing overhead

Consider first the matter of setting standards for materials; the best indicator of material usage per unit of product will be the consumption records of the previous period

This year:

A trolley consumed 11 meters of material

Next year:

Standard for next year at 10 meters

It would be pointless unless the company could demonstrate an ability to save one meter due to new processes or techniques

Standard for next year at 11 meters

It would be wasteful if this number included inefficiencies in selection of correct lengths and the consequent trimming of sizeable ends of metal which could not be used for any purpose other than sale for scrap

A firm needs to know precisely how much material it will require

Before Production

And

During the product's life cycle in production

Putting a Prototype into Production

Detailed Engineering Study

Considers:

- The complexity of design of the product
- The availability of the required quality of material
- The skill of labor required to make the product
- The engineering precision of the production machinery

The resultant figure will be the *ideal* or *engineered standard*

Amounts of material which should be required if every aspect of production works to maximum efficiency

But of course this seldom happens in reality where the engineered standards are rarely reached

Reasons behind this:

Machines break down

Machines require
recalibration

Workers make mistakes

Workers operate at faster
speeds than planned

Workers operate at a slower
rate than planned

Quality of raw materials
varies

Normal or Operating Standards

Amount of material which is expected to be achieved by a productive operation operating carefully and efficiently but which also reflect the normal perils which surround a process where so many things can go wrong

The materials' price standard will be based on external economic factors

Factors include:

Expected purchase pattern

Suppliers' estimates of price movements

Quantity discounts negotiated

Quality of raw material required

A business whose raw material inputs are subject to significant price fluctuations, e.g. commodities such as wheat, may reduce their budget periods to shorter terms

Labor usage and price standards are set in similar manner to those for materials

Detailed Engineering Study

Should be able to measure the time required by a skilled operative:

- To manufacture the product
- To assemble the product
- To test the product



Engineered Labor Usage Standards

Normal Standards would consider:

- The time taken for refreshment breaks
- Casual conversations with supervisors and colleagues
- fetching raw material and
- consumables and carrying finished components to the next workspace or collection point

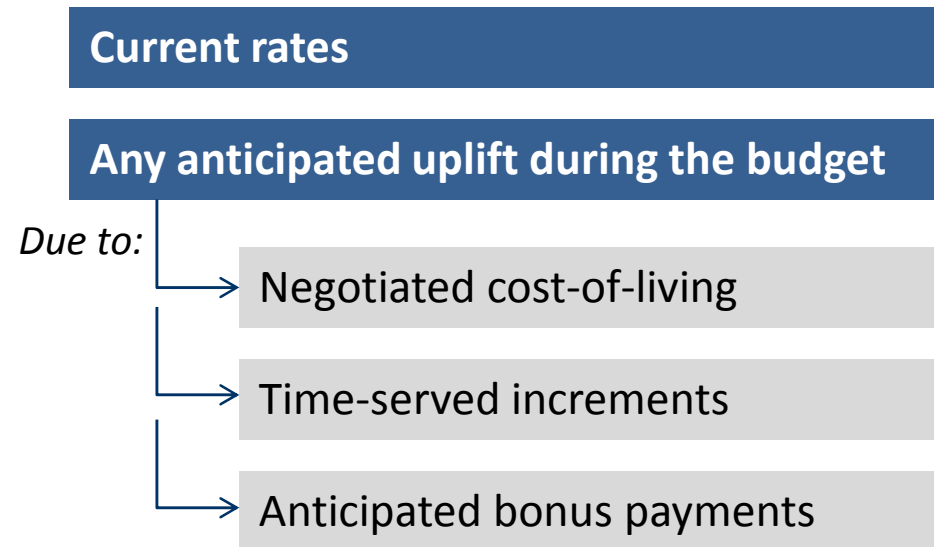


Normal Labor Usage Standards

Such a study would describe the quality of operative and the nature of training to be undertaken before production commences

While labor usage is usually measured in units of time, the standard price for direct labor input is measured in money value considering several factors

We have to consider:



Overhead standards tend to be determined by reference to one of the four standards already described

The standards for both variable and fixed overhead depend on three components:

The budgeted cost of each overhead

The allocation key to be used to allocate overhead to product

The volume expected of each allocation key to be used

Example

- Assume budgeted fixed overhead amounts to €100,000 for next year and the allocation key to be used is machine-hours for all fixed overhead. If 400,000 machine-hours is anticipated next year a budgeted fixed overhead rate of €0.25 per machine-hour will be used.
- A product that uses 10 machine-hours will have a standard fixed overhead rate of €2.50 for next year. A change in either (b) or (c) will have the effect of altering the standard overhead cost per unit even if (a) remains the same.

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The greatest role of budgets is in motivating the workforce to perform at their highest level of achievement

Budgets Should :

Be carrots to encourage and reward superior performance

But:

If the budget is slack, there is little incentive for an employee to sharpen performance (it has to be stretched but not slack)

The use of **normal standard**, provided there has been built into it an element of 'stretch' and rigor, will give the incentive to well-intentioned employees to strive to reach the level expected, and to reward them for so doing

Budgets Should NOT:

Be used as sticks to beat recalcitrant workers into shape

If employees are presented with a budget which is well beyond their reach they will not attempt to come anywhere near the level expected

The use of **ideal, engineered standards** is a short-sighted demotivational device which should be resisted

The setting of budgets and standards is therefore a task which involves every level of the organization and 'external observers' besides employees who know most about the processes under review

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Imagine a quarterly report on production cost performance along the following lines:

Third quarter ended 31 March 20x4

	Budget	Actual
Units produced	2 000	1 800
	€	€
Cost of materials	8 000	7 380
Cost of labour	6 000	5 310
Cost of overhead	2 000	2 050
Total production cost	16 000	14 740

The production facility has spent €1,260 less than budgeted but the production level was reduced from 2,000 to 1,800

Commentary by production manager to chief executive:

Yet again we've had a good quarter by coming in €1260 below budgeted cost. One more quarter like this and the production team will be wanting to share in the savings by way of a year-end bonus.

It is futile to compare the actual costs of making 1,800 with the budgeted costs of making 2,000; this could be dangerously misleading



But which column should be adjusted? Should we flex the budget column down to the 1,800 level or the actual column up to the 2,000 level?

The answer is that we must flex the budgeted column of costs down to the level of output actually achieved

A far more meaningful presentation would use a flexible budget presentation:

Third quarter ended 31 March 20x4

	Budget	Actual	Variance*
Planned production in units	2 000		
Actual production in units	1 800	1 800	
	€	€	€
Cost of materials	7 200	7 380	(180)
Cost of labour	5 400	5 310	90
Cost of overhead**	1 900	2 050	(150)
	14 500	14 740	(240)

* The term 'variance' means the difference between the budgeted (or standard) cost and actual cost.

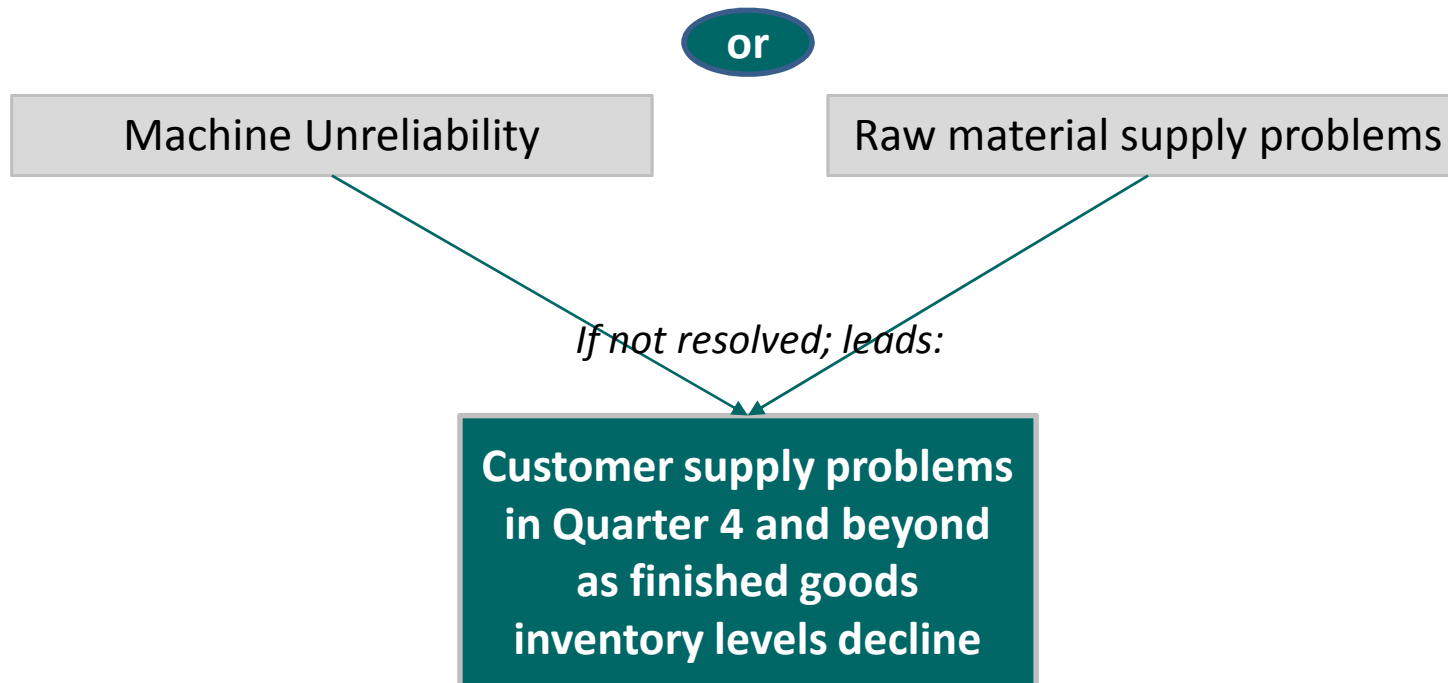
** Overhead comprises a fixed component of €1000 and a variable component of €0.50 per unit produced
(1800 × €0.50 = €900)

Net effect of these overruns & underruns was €240, or 1.65% over budget

This report now gives a better picture of the activities of production over the third quarter to 31 March 20x4; is much more informative than the earlier one

But even in the preferred report, an explanation of why the production level had fallen from 2,000 to 1,800 would be necessary

This variance may reflect:



Note that the flexible budget for 1,800 units was based on the standard cost of the unit.

	€
Material	4.00
Labour	3.00
Variable overhead	0.50
Fixed overhead*	0.50
	<u>8.00</u>

* Based on the planned production level of 2000 units per quarter.

The per-unit fixed overhead component of standard cost is only valid if 2,000 units are made, simply because the fixed overheads amount to €1,000 regardless of volume produced

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The variances column in the flexible budget is informative but only to a degree

Take materials: for management of the production facility to be told that €180 more was spent in Quarter 3 than anticipated tells them that something did not go according to plan, but gives them no indication of where the problem is. More detail is required.

Material costs can be more (or less) than standard for only two reasons:

- (a) actual production used more (or less) material than planned, and/or
- (b) the price to purchase the material was more (or less) than planned.

A combination of these two reasons leads us to split them out for individual analysis:

$$\begin{aligned}\text{Material efficiency variance} &= (\text{Standard quantity} - \text{Actual quantity}) \times \\ &\quad (\text{Standard price per unit price}) \\ &= (SQ - AQ) SP\end{aligned}$$

$$\begin{aligned}\text{Material price variance} &= (\text{Standard price per unit} - \text{Actual price per unit}) \times \\ &\quad (\text{Actual quantity used}) \\ &= (SP - AP) AQ\end{aligned}$$

Example 1

A kilt-making firm in London makes 50 kilts per month. Last month, against a budgeted production cost of €3,000 it reported a production cost of €2,860. Management has called for an explanation. You determine that the standard cost of each kilt contains four meters of tartan tweed purchased in bales from a company in the Outer Hebrides, each meter costed at €15. Last month each kilt consumed 4.4 meters and each meter cost €13.

Application of the above formulae produces the following variances:

$$\begin{aligned}\text{Material efficiency variance} &= (SQ - AQ) SP \\ &= [(50 \times 4) - (50 \times 4.4)] €15 \\ &= (200 - 220) €15 \\ &= €300 \text{ adverse}\end{aligned}$$

$$\begin{aligned}\text{Material price variance} &= (SP - AP) AQ \\ &= (€15 - €13) 220 \\ &= €440 \text{ favourable}\end{aligned}$$

$$\begin{aligned}\text{Total material variances} &= €300 \text{ adverse} + €440 \text{ favourable} \\ &= €140 \text{ net favourable}\end{aligned}$$

Here we learnt two new terms: adverse variance and favorable variance

Adverse variance

Variance where actual cost is above
standard cost

Favorable variance

Variance where actual cost is less
than standard cost

Calculation of variances is of little value unless it prompts investigation into possible reasons for them. Here are a number of possible reasons for the variances so far calculated:

Material efficiency = €300 adverse

- faulty bale of tweed which prevented easy matching of pattern;
- inexperienced staff leading to waste, mismatch or excessive hems;
- use of sewing machines which require maintenance;
- pilferage among staff leading to inventory losses;
- poor-quality material (see below).

Material price = €440 favourable

- change of supplier (although this looks unlikely given the facts supplied);
- quantity discounts given being more generous than planned for;
- inferior quality of tweed (which leads to more metres being used);
- end-of-tartan bales which are supplied at a lower price.

Labour cost variances are caused by a combination of the same two reasons:

- (a) actual production requiring more (or less) time than planned; and/or
- (b) the labour rates actually paid were more (or less) than planned.

A combination of these two reasons leads us to split them out for individual analysis:

$$\begin{aligned}\text{Labour efficiency variance} &= (\text{Standard time allowed} - \text{Actual time taken}) \times \\ &\quad (\text{Standard rate per hour}) \\ &= (ST - AT) SR\end{aligned}$$

$$\begin{aligned}\text{Labour rate variance} &= (\text{Standard rate per hour} - \text{Actual rate per hour}) \times \\ &\quad (\text{Actual time taken}) \\ &= (SR - AR) AT\end{aligned}$$

Note that the term 'rate' is used here for labor variances and 'price' is used for material. Sometimes 'price' is used for both variances. Also, the formulae are sometimes laid out as follows: e.g. Labor rate variances $AT(SR - AR)$.

Example 2

At the end of the same month as above, the kilt makers' production department reported its actual labor costs equaling the budgeted labor costs at €3,000. Initially management is pleased with this outcome until you investigate the standard costs and produce the variances which follow. Each kilt's standard time to cut and sew is three hours, each hour being paid at the standard rate of €20. In the month under review 175 hours were used, costing €2,999.50.

Application of the above formulae produces the following variances:

$$\begin{aligned}\text{Labour efficiency variance} &= (ST - AT) SR \\ &= [(50 \times 3) - (50 \times 3.5)] 20 \\ &= (150 - 175) 20 \\ &= €500 \text{ adverse}\end{aligned}$$

$$\begin{aligned}\text{Labour rate variance} &= (SR - AR) AT \\ &= (20 - 17.14) 175 \\ &= €500 \text{ favourable}\end{aligned}$$

$$\begin{aligned}\text{Total labour variances} &= €500 \text{ adverse} + €500 \text{ favourable} \\ &= \text{Nil}\end{aligned}$$

Example 2

Management would be well advised to ignore the fact that labor variances net to Nil because both efficiency and rate require investigation and possible managerial remedy action. Possible explanations?

Labour efficiency = €500 adverse

- use of faulty material requires more time spent per kilt;
- uncertain supply of material leading to broken runs and downtime;
- inexperienced staff;
- untrained staff or faulty equipment.

Labour rate = €500 favourable

- lower calibre of staff used during the month leading to more time required;
- anticipated pay settlement deferred until later in the year;
- less overtime paid than budgeted for.

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As in the preparation and monitoring of budgets it is imperative that personnel are not judged, or held responsible, for variances which lie outside their control

Do-it-Yourself Example in Materials and Labor Variances

Consider the following standard cost data for a business making cases for carriage clocks:

	€
Direct material 0.4 sq m metal @ €20 per sq m	8.00
Direct labour 2 hours @ €10 per hour	20.00
	28.00
Actual statistics for last month were:	
Cases produced	6 000
Square metres of metal purchased @ €22 per sq m	2 100 sq m
Square metres of metal used	2 280 sq m
Direct labour @ €10.10 per hour	11 100 hours

Required

Using the pro forma analysis sheet below, calculate the material and labor variances and consider possible reasons for these variances. The material variances should recognize the responsibilities of the production manager and purchasing manager respectively.

Material variances		Labour variances	
Material efficiency	$= (SQ - AQ)SP$	Labour efficiency	$= (ST - AT)SR$
	=		=
	=		=
	=		=
Possible explanations?			
Material price	$= (SP - AP)AQ$	Labour rate	$= (SR - AR)AT$
	=		=
	=		=

Worked solution

Material efficiency

$$\begin{aligned} & - (SQ - AQ)SP \\ & = [(6000 \times 0.4) - (6000 \times 0.38)] \text{€}20 \end{aligned}$$

$$= (2400 - 2280) \text{€}20$$

$$= \text{€}2400 \text{ favourable}$$

- better calibrated cutting equipment;
- use of more appropriate sheets of metal;
- superior quality of metal;
- change in design since setting of standard.

Labour efficiency

$$\begin{aligned} & - (ST - AT)SR \\ & = [(6000 \times 2) - (6000 \times 1.85)] \text{€}10 \end{aligned}$$

$$= (12000 - 11100) \text{€}10$$

$$= \text{€}9000 \text{ favourable}$$

- greater motivation;
- result of increased training;
- improved mechanisation.

Worked solution (Continued)

$$\begin{aligned}\text{Material price} &= (SP - AP)AQ \\ &= (\text{€}20 - \text{€}22)2100 \\ &= \text{€}4200 \text{ adverse}\end{aligned}$$

- price rise due to world shortage;
- change of supplier.

$$\begin{aligned}\text{Labour rate} &= (SR - AR)AT \\ &= (\text{€}10 - \text{€}10.10) 11\,100 \\ &= \text{€}1110 \text{ adverse}\end{aligned}$$

- unforeseen wages settlement;
- more overtime;
- superior level of operatives.

Note 1

Post:	Responsible For:	Units used:
Production Manager	Volume of raw material used	The square meters of metal drawn from stores into production
Purchasing Manager	Price obtained for the quantity purchased	Price variance's 'actual quantity' is deemed to be the number of square meters purchased

Had the production manager also been held responsible for the material price variance, the variance would have been:

$$\begin{aligned} & (SP - AP) AQ \\ &= (\text{€}20 - \text{€}22) 2280 \\ &= \text{€}4560 \text{ adverse} \end{aligned}$$

Note 2: Variances should be calculated at the stage when they arise and the products should be passed on to the next department or process at standard cost

Example

- The carriage clock casings will be passed to the assembly department; it would be inappropriate to 'pass on' the €2400 favorable material efficiency variance arising in the casing department.
- This variance is the responsibility of the casing department's manager.

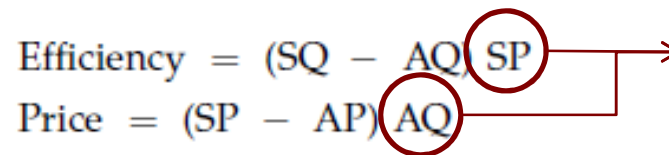
Note 3: Consistent variances arising in the same area of operations may indicate that the standard is no longer appropriate and needs adjustment

Example

- In the casing department, it may be seen that a sizeable favorable variance in labor efficiency occurs every month. If this is so then the labor input process needs further engineering study to establish the new standard time in the light of improved machining and automated techniques.
- A detailed study could conclude that 1.85 hours per clock actually achieved above is excessive in comparison with what could be achieved at normal efficiency.

Note 4: We caution readers against memorizing the formulae without fully understanding the concepts behind them

In our experience managers understand the identities inside the brackets but are unsure of those outside the brackets. Take the material variances:

$$\begin{aligned}\text{Efficiency} &= (\text{SQ} - \text{AQ}) \text{SP} \\ \text{Price} &= (\text{SP} - \text{AP}) \text{AQ}\end{aligned}$$


Why do we use standard price outside the brackets with efficiency and actual quantity outside the brackets with price?

Careful thought about what is being examined will give a satisfactory answer to this question

In the efficiency variance formula

- We analyze the **financial impact** on the business of the manufacturing or assembly process taking more or less material than standard to do the job
- The physical quantity of material beyond standard is valued at standard price, otherwise the combined sum would be a hotch-potch of two variances, efficiency and price.
- By holding all prices at standard, management can gain an insight into the efficiency of production.

In the price variance formula

- The gain or slippage in price achieved for raw materials used or purchased is multiplied by the actual quantity used or purchased.
- Were we to use the standard quantity we would produce a figure which bore no resemblance to reality; thus, in the worked example above the purchasing department actually purchased 2100 square meters of metal, each one €2 more than standard. If the standard quantity of 2400 square meters were used, the purchasing manager would be penalized with a higher adverse variance than he incurred.

In the example we could argue that he kept the purchasing of metal to a minimum so as to minimize the impact of the increase in unit price; The price variance recognizes this foresight by using actual quantity outside the brackets

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Accountants use an allocation key to spread overhead over the products in a manner which best reflects the products' usage of the overhead resources

Variance analysis

Is the examination of the difference between how much overhead was incurred and how much should have been incurred

Is based on the allocation keys used to spread overhead

In our variable overhead analysis, we can use efficiency and price as the criteria for our analysis of over- or under-spending

Example

Consider the carriage clock business again (see section 13.6.1). Imagine that at the beginning of the year the management of the company planned to make 60,000 casings and that each casing would take two hours to make. Variable overheads for 120 000 direct labor hours were budgeted at €180,000, or €1.50 per direct labor hour, or €3 per unit.

In the month under review variable overheads amounted to €17,200 when 6,000 cases were manufactured and 11,100 direct labor hours were recorded. Variable overhead analysis starts with the simple question: how much variable overhead should have been incurred, given that 6,000 units were produced, and how much was incurred?

Standard cost of variable overheads	Less	Actual cost of variable overheads
(6000 × 2 hours × €1.50)		
= €18 000	—	€17 200
= €800 favourable		

Example (Continued)

For many managements the level of analysis of this variance would be sufficient. But it conceals two components which merit further examination. Because the business is employing direct labor hours as an allocation key, we can break down this overall variance of €800 into efficiency and price (usually referred to as spending).

Variable overhead

Efficiency variance

= Standard cost of flexible budget time allowance for units produced	Less Standard cost of actual time taken for units produced
= $(6000 \times 2 \times \text{€}1.50)$	– $(11\,100 \times \text{€}1.50)$
= €18 000	– €16 650
<u>€1350 favourable</u>	

Variable overhead

Spending variance

= Standard cost of actual time taken for units produced	Less Actual costs incurred
= $11\,100 \times \text{€}1.50$	
= €16 650	– €17 200
= €550 adverse	

Should not surprise management given that direct labor hours is the allocation key for variable overhead and that a favorable efficiency variance for direct labor has already been calculated

Of course this is highly expected if the allocation key (in this case direct labor hours) is directly correlated to the components comprising variable overheads –fewer direct labor hours leads to fewer variable overheads being incurred

Example (Continued)

The variable overhead spending variance is potentially a more interesting variance than the efficiency because:

Management

Instead of incurring €16,650 in variable overhead

The level of overhead allowed for in the flexible budget given the actual number of direct labor hours incurred

The business spent €17,200, or €550 too much

↓
Caused by:

- Either an increase in prices paid for the components comprising variable overhead
- Or poor control (inefficiency!) over the use of these items

Failure to go behind the single €800 favorable variable overhead variance would have concealed from management a serious overspending on some items making up variable overhead

Consider next fixed overheads

Example

We now provide further extension of the example portraying the standard costing system of the casing department of the carriage clock business (see section 13.6.1): the budgeted fixed overheads for the year were €360,000 to be incurred evenly through the year, i.e. €30,000 per month. Actual fixed overhead amounted to €32,500. Fixed overhead is to be allocated using direct labor hours. Management expect 120,000 direct labor hours to be incurred annually. Therefore the predetermined fixed overhead rate for this year is:

$$\frac{€360\,000}{120\,000 \text{ hours}} = €3 \text{ per direct labour hour}$$

If, during the month under review, exactly 5000 units were produced (leading to 10 000 standard direct labor hours being incurred) and €30 000 was the actual level of monthly spend on fixed overhead, then no fixed overhead variances would arise. This seldom happens!

$$\frac{€30\,000}{10\,000 \text{ hours}} = €3 \text{ per direct labour hour}$$

Example (Continued)

Again the straightforward analytical question would be: how much fixed overhead was incurred against how much was budgeted for?

Budgeted amount	Less	Actual amount
€30 000	—	€32 500
= €500 adverse		

But this analysis, usually termed the fixed overhead spending variance, does not reveal the full picture (unlike the total variance of €800 calculated for variable overhead). True it yet again leads management to investigate the apparent cost overruns in the elements comprising fixed overheads (e.g. an increased outlay for factory rent or unforeseen recruitment in supervisory services). But it ignores the fact that more fixed overheads were applied to the product than were budgeted for. We must therefore calculate another variance, again using the budgeted amount of €30,000 as one of the elements:

Fixed overhead		Amount applied to
denominator variance	= Budgeted amount	Less units produced
	= €30 000	— (6 000 × 2 × €3)
	= €30 000	— €36 000
	= €6 000 favourable	

Here we see fixed overhead appearing to behave like variable overheads, but we have to take care of appearances!!

At the beginning of the budget period, a predetermined overhead allocation rate of €3 per direct labor hour was calculated

If everything went exactly according to plan, i.e. 60,000 cases were produced consuming 120,000 direct labor hours, there would be no denominator variance because the actual and budgeted denominator would be the same



But **when the actual volume of output used in the denominator moves away from budget**, units of production are allocated more or less fixed overhead than budgeted for.

In our example, we can see a favorable variance taking place

Each casing uses two direct labor hours, therefore 2,000 more direct labor hours were recorded than planned.

If each recorded hour is used to allocate fixed overhead then $2000 \times \text{€}3 = \text{€}6,000$ more fixed overhead has been allocated to production than was budgeted for.



This is deemed to be a favorable variance because volume of output is greater than budgeted

Do-it-Yourself Example in Variable and Fixed Overheads

A company uses a standard cost system to plan and control its manufacturing process of compact discs. The standard cost of a disc, based on a denominator volume of one million discs per annum, includes four machine-hours of variable overhead at €0.50 per hour and four machine-hours of fixed overhead at €10 per hour. Actual output for the year under review was 1.2 million discs; actual variable overhead was €1,900,000; actual machine-hours recorded were 4,100,000; actual fixed overhead was €39,000,000.

Variable overhead variances

$$\begin{aligned}\text{Efficiency} &= [\text{Standard cost of flexible budget time allowance for units produced}] && \text{Less } [\text{Standard cost of actual time taken for units produced}] \\ &= \\ &= \\ &= \end{aligned}$$

Possible explanation?

$$\begin{aligned}\text{Spending} &= [\text{Standard cost of actual time taken for units produced}] && \text{Less } [\text{Actual costs incurred}] \\ &= \\ &= \\ &= \end{aligned}$$

Possible explanation?

Do-it-Yourself Example in Variable and Fixed Overheads (Continued)

Fixed Overhead Variances

Predetermined fixed overhead application rate

$$= \frac{\text{€}}{\text{machine-hours}} = \text{€10 per machine-hour}$$

Spending = Budgeted amount *Less* Actual amount

=

=

Possible explanation?

Denominator = Budgeted amount *Less* Amount applied to units produced

=

=

=

Possible explanation?

Do-it-Yourself Example in Variable and Fixed Overheads (Solution)

Variable overhead variances

Efficiency = [Standard cost of flexible budget time allowance for units produced]	<i>Less</i>	[Standard cost of actual time taken for units produced]
= (1 200 000 × 4 × €0.50)	–	(4 100 000 × €0.50)
= €2 400 000	–	€2 050 000
= €350 000 favourable		

Possible explanation? The number of hours budgeted for downtime did not materialise and therefore the machine-hours recorded in manufacturing 1.2 million compact discs were 700 000 less than anticipated. Alternatively, the standard time of four machine-hours per disc needs to be tightened up.

Spending = Standard cost of actual time taken for units produced	<i>Less</i>	Actual costs incurred
= (4 100 000 × €0.50)	–	€1 900 000
= €2 050 000	–	€1 900 000
= €150 000 favourable		

Possible explanation? Given the variable overhead allowed for every machine-hour recorded, the company was entitled to spend €2 050 000 on variable overhead; instead it spent €1 900 000. This could indicate tighter operational control over the components of variable overhead or more advantageous prices negotiated than budgeted for.

Do-it-Yourself Example in Variable and Fixed Overheads (Solution)

Fixed Overhead Variances

Predetermined overhead rate

$$= \frac{€40\,000\,000}{4\,000\,000} \text{ machine-hours} = €10 \text{ per machine-hour}$$

Spending = Budgeted amount	<i>Less</i>	Actual amount
= €40 000 000	–	€39 000 000
= €1 million favourable		

Possible explanation? The sheer magnitude of fixed overheads may be surprising. But in high-technology industry with a high level of automation and high level of research and development expenditure, this proportion between variable and fixed overhead can be expected. The difference between the two amounts is small in percentage terms (2.5 per cent) and would be perhaps due to slightly less expensive machines being acquired, thereby lowering the fixed annual depreciation charge or to an extended view on machine life cycles (which would have a similar effect on depreciation).

Denominator = Budgeted amount	<i>Less</i>	Amount applied to units produced
= €40 000 000	–	1 200 000 × 4 × €10
= €40 000 000	–	€48 000 000
= €8 000 favourable		

Possible explanation? Production was running at 20 per cent higher than budgeted. This inevitably led to 20 per cent more fixed overhead being *applied* to the production than budgeted. (Note: the increased production would have no impact on the amount of fixed overhead actually *spent*.) If this increase in output is viewed as normal then the denominator volume has to be increased for next year's predetermined overhead rate thereby reducing the cost of each unit of product.

Agenda

13.1 Introduction

13.2 Setting Standards

13.3 A Word about Motivation

13.4 Flexible Budgets

13.5 The Anatomy of Variances: Materials and Labor

13.6 Responsibility for Variances

13.7 Variable and Fixed Overhead Analysis

13.8 Investigation of Variances

13.9 Sales Variances

Review Questions

Case Studies

Good managers are busy people; they do not want to be pestered by sheets of detailed numbers revealing that everything is under control

Exception Reporting or Management by Exception

Well-presented report revealing the most significant variances only, not the workings behind the variances, only the big numbers

But defining 'out of control' is not easy. The limits must be determined over time and through experience, and in consultation with the engineering staff

**Determining
a percentage limit is a
useful start**

For example, all material variances
below 4 per cent are not reported



But a rigid interpretation of this limit may lead to important control information being suppressed. A business may consistently report a 3.8 per cent adverse variance on material efficiency, month after month, which should provoke an investigation into raw material usage but because the monthly percentage falls below the limit, the cumulative impact of this variance escapes managerial attention

Another difficulty in investigating variances is the interdependencies among the variances; a decision to investigate one variance may lead to additional work on others

A selection of possible linkages:

1

An adverse materials efficiency variance may be caused by the use of inferior quality materials which would be reflected in favorable material price variance.

2

The same inferior-quality raw materials may lead to an adverse labor efficiency variance because more time has to be spent selecting suitable batches of material.

3

A favorable labor rate variance caused by employing less-skilled operatives may be counterbalanced by adverse variances in labor and material efficiency.

4

If direct labor is used to apply variable overhead then the variable overhead efficiency variance will be of the same magnitude as the labor efficiency variance even although there may exist a tenuous link between labor efficiency and the consumption of resources comprising variable overhead.

5

A favorable denominator variance in fixed overhead may provoke the production staff to aim for even higher levels. But this would have a negative impact on inventory levels (high) and cash levels (low).

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Review Questions

Case Studies

Case

The manager of a sports goods chain of shops is delighted at the monthly management accounting report for the month of May. From the information contained therein he learns that his contribution margin on tennis shoes rose by €25,000 from the budgeted level of €85,000 even though he is aware that volume of sales has fallen. He sells two varieties, a top-of-the-range pair, Ace'em, selling for €50, and a cheaper version for less athletic individuals, Plodder, selling for €12.50. The budgeted and actual figures for May are given below (a pair of shoes = 1 unit).

Budgeted figures

	Ace'em			Plodder			Total	
	Units	Price	Total	Units	Price	Total	Price	Total
Sales	2000	€50.00	€100 000	18 000	€12.50	€225 000	€16.25	€325 000
Bought-in cost	2000	30.00	60 000	18 000	10.00	180 000	12.00	240 000
Contribution margin	2000	€20.00	€40 000	18 000	€2.50	€45 000	€4.25	€85 000

Case (Continued)

Note that the total columns comprise the additions of the units sold and revenue and bought-in costs for both styles of shoes; the total 'price' is derived by dividing the total figures of revenue and bought-in costs by 20 000. The figures of €16.25, €12.00 and €4.25 are therefore weighted average numbers (heavily influenced by the preponderance of the cheaper Plodder shoe).

Actual results

	Ace'em			Plodder			Total	
	Price	Total	Units	Price	Total	Units	Price	Total
Sales	€51.25	€205 000	10 000	€12.50	€125 000	14 000	€23.57	€330 000
Bought-in cost	30.00	120 000	10 000	10.00	100 000	14 000	15.71	220 000
Contribution margin	€21.25	€85 000	10 000	€2.50	€25 000	14 000	€7.86	€110 000

Case (Continued)

He asks for your help in analyzing the increase in contribution of €25 000 in the face of a decline in sales volume. An initial glance at the numbers in the case shows that the high-ticket item, Ace'em, has doubled in sales volume and he has managed to squeeze an extra €1.25 per pair from the customers. The very significant positive impact of this increase in contribution from Ace'em is reduced by a serious decline in the Plodder line. The manager believes that this drop is due to an increase in consumer spending power with a consequent increase in leisure outlays. Even the least promising tennis player is prepared to spend money on upmarket clothes and equipment. A more detailed calculation of line contributions reveals the following pattern in contribution margin percentages (contribution margin as a percentage of sales price).

	Ace'em	Plodder	Total
Budget	40%	20%	26.154%
Actual	41.463%	20%	33.347%